

ITAO7106 Marketing Analytics

Supplementary Assessment

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PART 1 — Customer Segmentation and Targeting

1. Introduction

Companies entering the US market face significant challenges identifying commercially attractive geographic markets due to the substantial heterogeneity of the US urban population across racial composition, age, income, and unemployment. A uniform marketing strategy is unlikely to be effective across all cities. Customer segmentation — grouping markets based on shared characteristics — is widely recognised as a foundational capability for targeted marketing strategy (Wedel and Kamakura, 2000). This report presents a data-driven segmentation of 49 major US cities using cluster analysis applied to six demographic variables, validated through Linear Discriminant Analysis (LDA). A target segment is identified and justified on both statistical and commercial grounds, with strategic recommendations for Place and Promotion derived directly from the analytical findings.

The approach addresses the customer heterogeneity problem: treating all markets as homogeneous leads to inefficient resource allocation and misaligned messaging. As Hair et al. (2019) note, cluster analysis is particularly suited to identifying natural groupings in multivariate data where no prior classification scheme exists. LDA complements this by providing statistical validation of group separation and enabling probabilistic segment assignment (McLachlan, 1992).

2. Methodology

2.1 Data and Preprocessing

The dataset contains demographic information for 49 US cities across six variables: percentage Black population (% Black), percentage Hispanic population (% Hispanic), percentage Asian population (% Asian), median age, unemployment rate, and per capita income (\$000s). Data was provided in pre-standardised z-score form, ensuring all variables contribute equally to distance calculations regardless of original measurement scales (Hair et al., 2019). Standardisation is essential because clustering

algorithms such as k-means are sensitive to scale differences — income measured in thousands would otherwise dominate unemployment measured as a percentage.

Exploratory data analysis (EDA) was conducted prior to clustering. A correlation matrix (Figure 1) was computed across all six raw demographic variables. Distribution analysis via histograms (Figure 2) revealed that per capita income peaks around \$20,000, unemployment is right-skewed with most cities at 5–7%, and % Black varies widely from near 0% to 76%, indicating substantial inter-city variation. Key correlations include % Black and unemployment ($r = 0.49$), unemployment and income ($r = -0.48$), and median age and income ($r = 0.50$), confirming meaningful interrelationships between segmentation variables that justify their combined use.

Additionally, a Power BI dashboard was developed to visualise cluster distributions geographically and demographically across the 49 US cities, providing an interactive complement to the R-based analysis (see Appendix, Figure 22).

Correlation Matrix — Demographic Variables

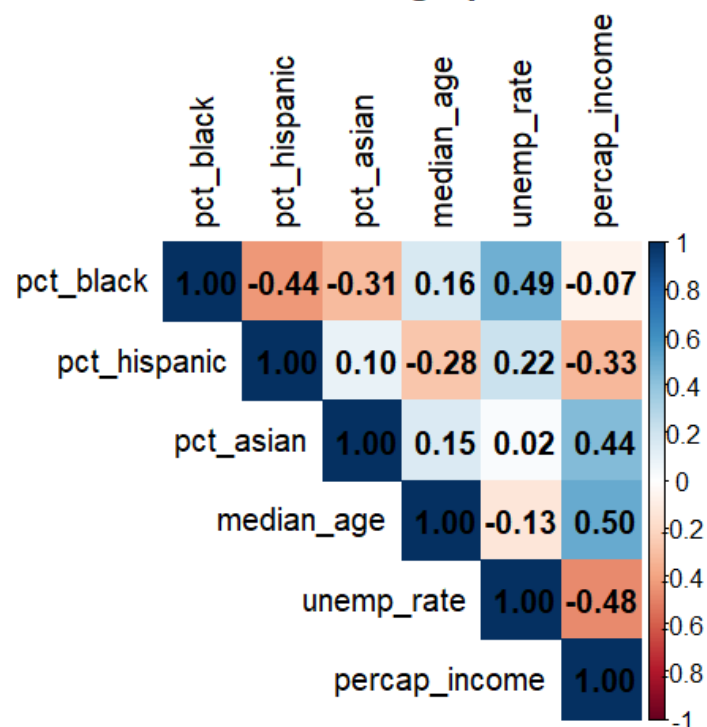


Figure 1. Correlation Matrix — Demographic Variables

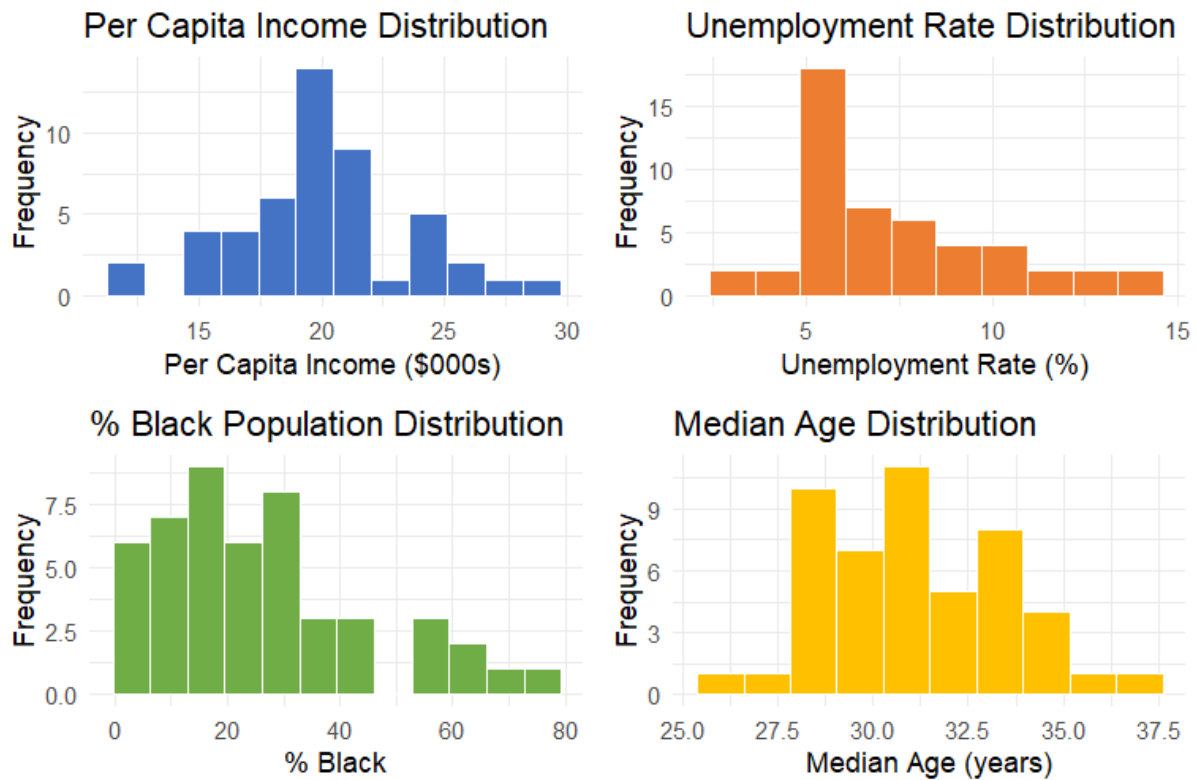


Figure 2. Distribution of Key Demographic Variables

2.2 Cluster Analysis

K-means clustering was applied to the standardised z-score variables. K-means was selected for its computational efficiency, scalability, and wide application in marketing segmentation contexts (Jain, 2010). The algorithm partitions observations by minimising within-cluster sum of squares (WSS), ensuring cities within each segment are more similar to each other than to cities in other segments.

The optimal number of clusters was determined using two complementary methods. The elbow method (Figure 3) shows WSS decreasing from 316 at $k=1$ to 199 at $k=3$, with diminishing returns beyond this point, indicating $k=3$ as suitable. The silhouette method (Figure 4) yields an average silhouette width of 0.19 at $k=3$. Although the silhouette technically peaks at $k=9$, a nine-cluster solution is impractical for marketing decision-making and lacks managerial interpretability. A three-cluster solution provides a balance between statistical quality and actionable segmentation, consistent with best practice in applied marketing analytics (Wedel and Kamakura, 2000).

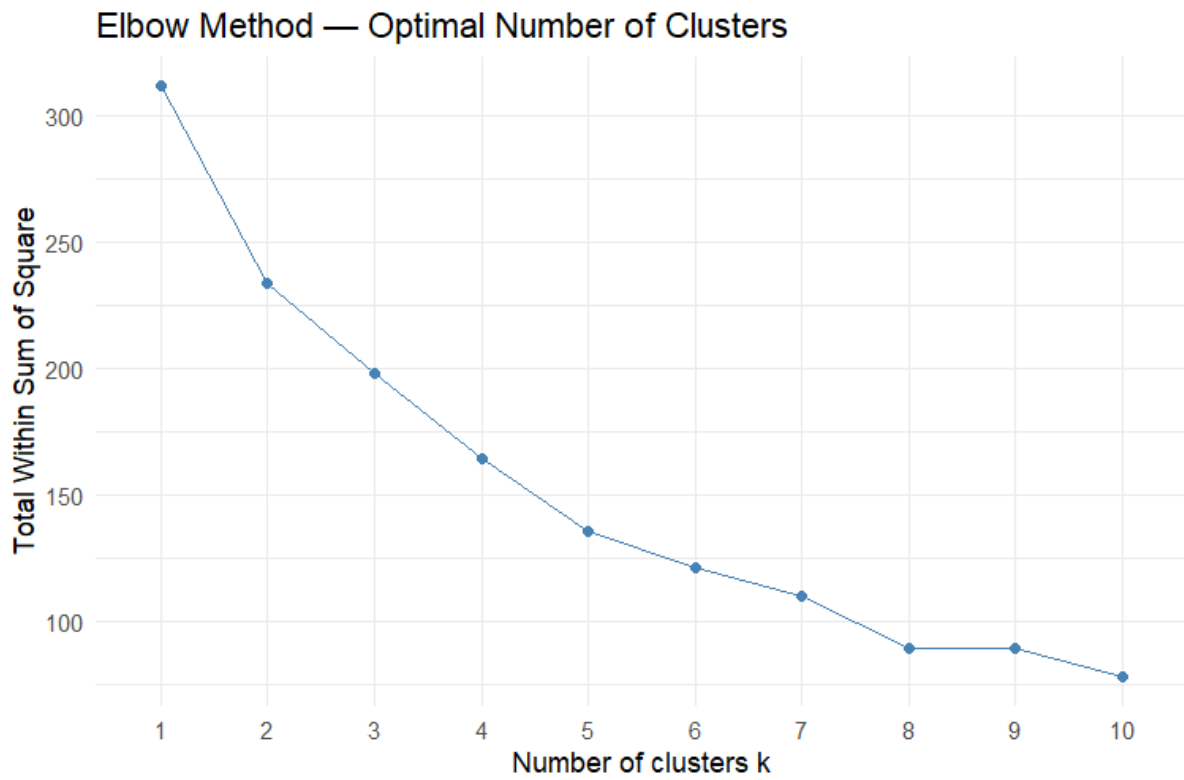


Figure 3. Elbow Method — Optimal Number of Clusters

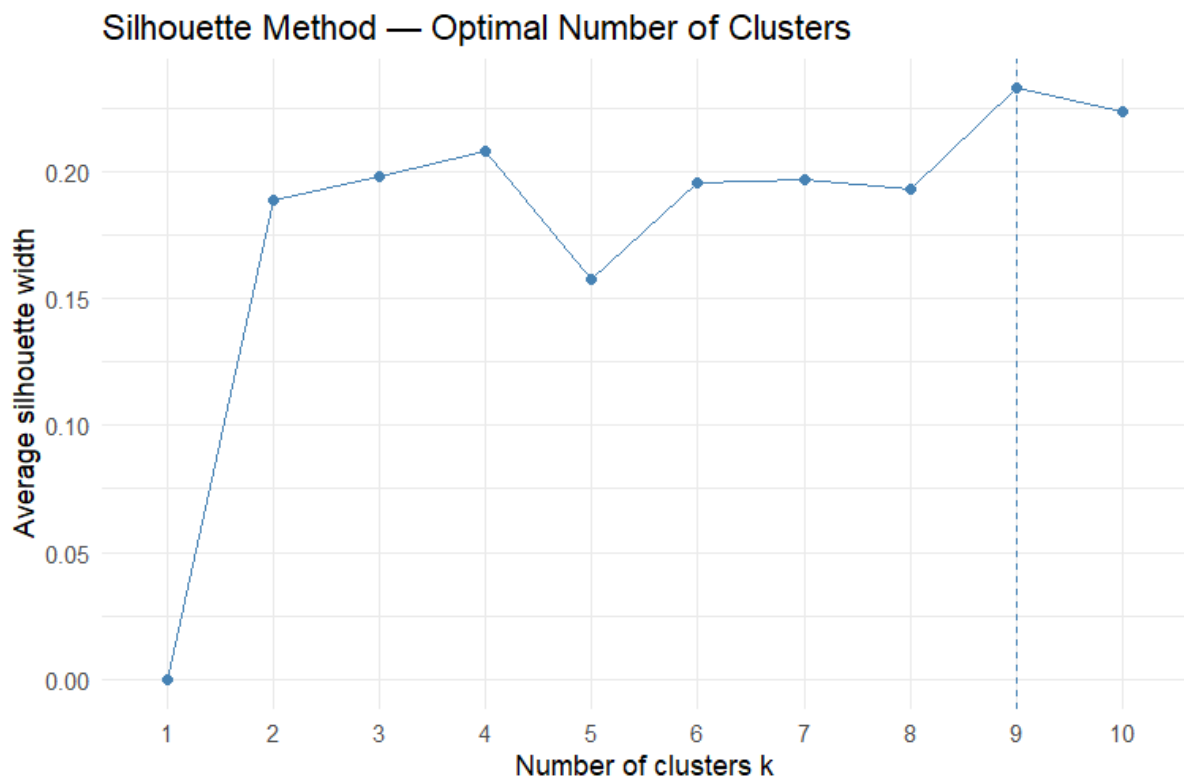


Figure 4. Silhouette Method — Optimal Number of Clusters

2.3 Linear Discriminant Analysis

LDA was applied following cluster assignment to validate the statistical separability of identified segments and assess classification accuracy. LDA identifies linear combinations of the original demographic variables that maximally discriminate between cluster groups (McLachlan, 1992). The dataset was split 70/30 into training (34 cities) and test sets (15 cities) using stratified random sampling, with equal prior probabilities assigned across three clusters. ANOVA F-tests were additionally computed for each demographic variable to confirm significant between-cluster differences, providing statistical justification for the segmentation solution.

3. Results and Discussion

3.1 Cluster Profiles

K-means identified three distinct city segments. Table 1 presents the average demographic profile for each cluster.

Cluster	N	% Black	% Hispanic	% Asian	Med. Age	Unemp %	Income \$k
Cluster 1	28	29.4	7.0	3.8	32.0	6.3	21.5
Cluster 2	15	11.1	34.0	6.0	30.1	7.1	18.9
Cluster 3	6	50.5	14.7	5.2	29.5	12.3	15.8

Table 1. Cluster Demographic Profile Summary

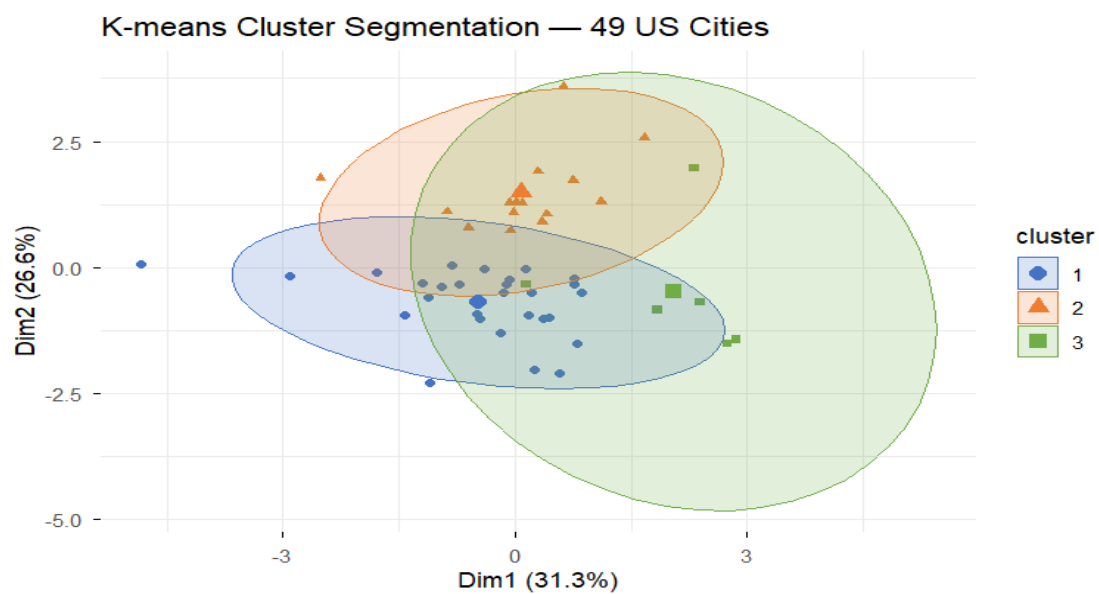


Figure 5. K-means Cluster Segmentation — 49 US Cities

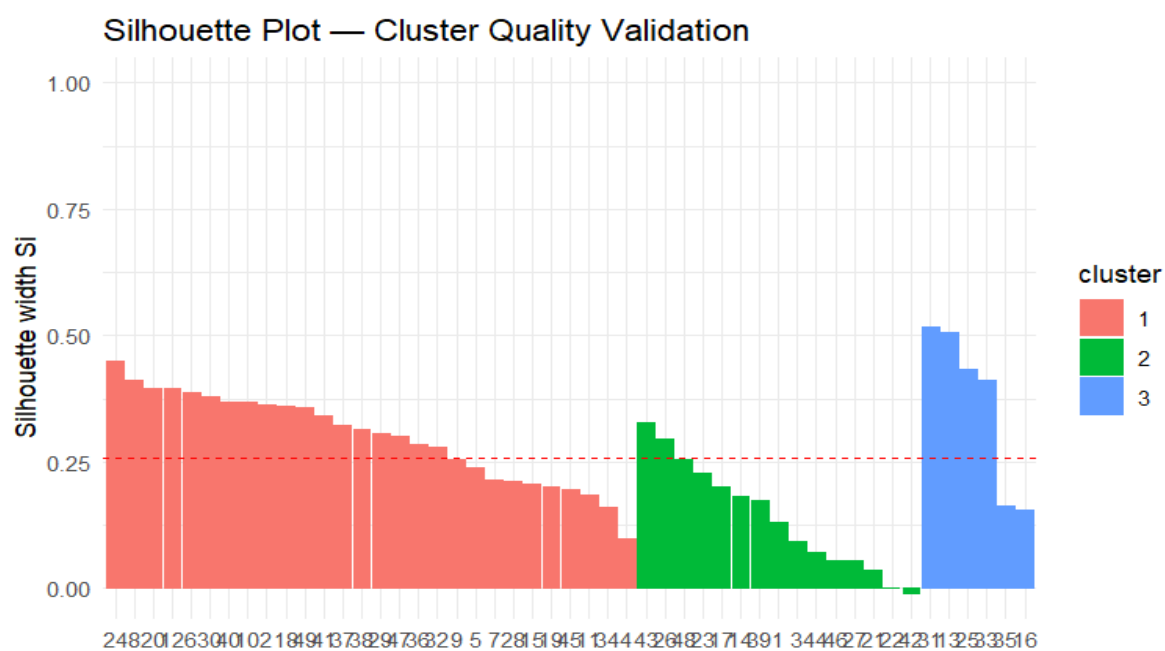


Figure 6. Silhouette Plot — Cluster Quality Validation

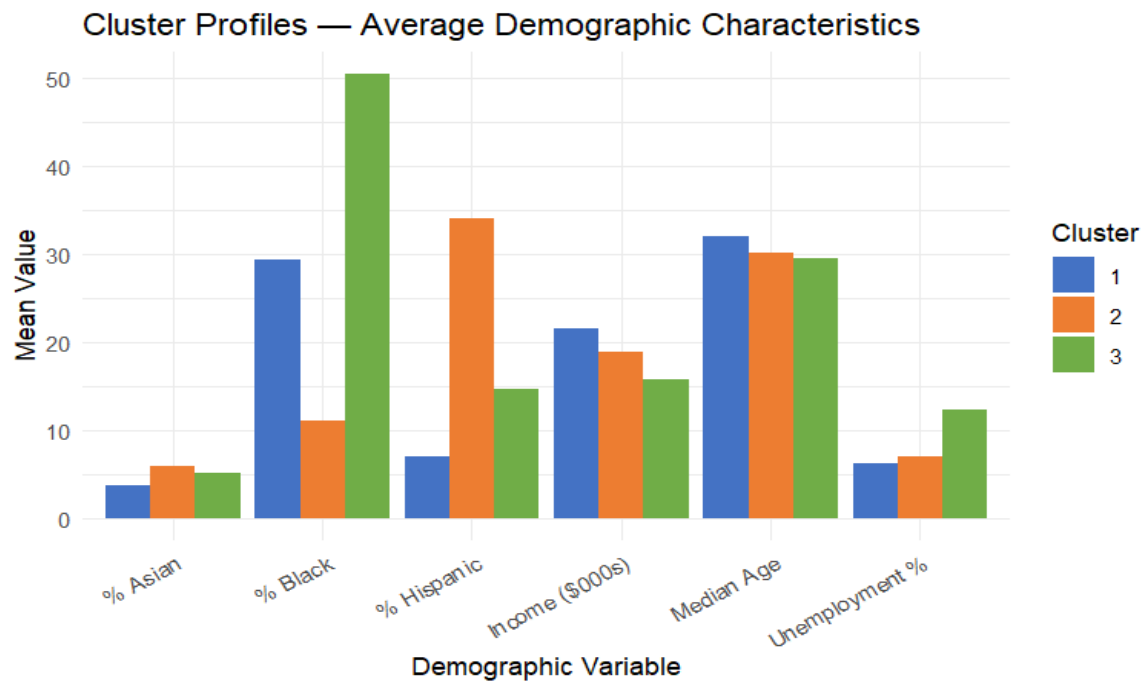


Figure 7. Cluster Profiles — Average Demographic Characteristics

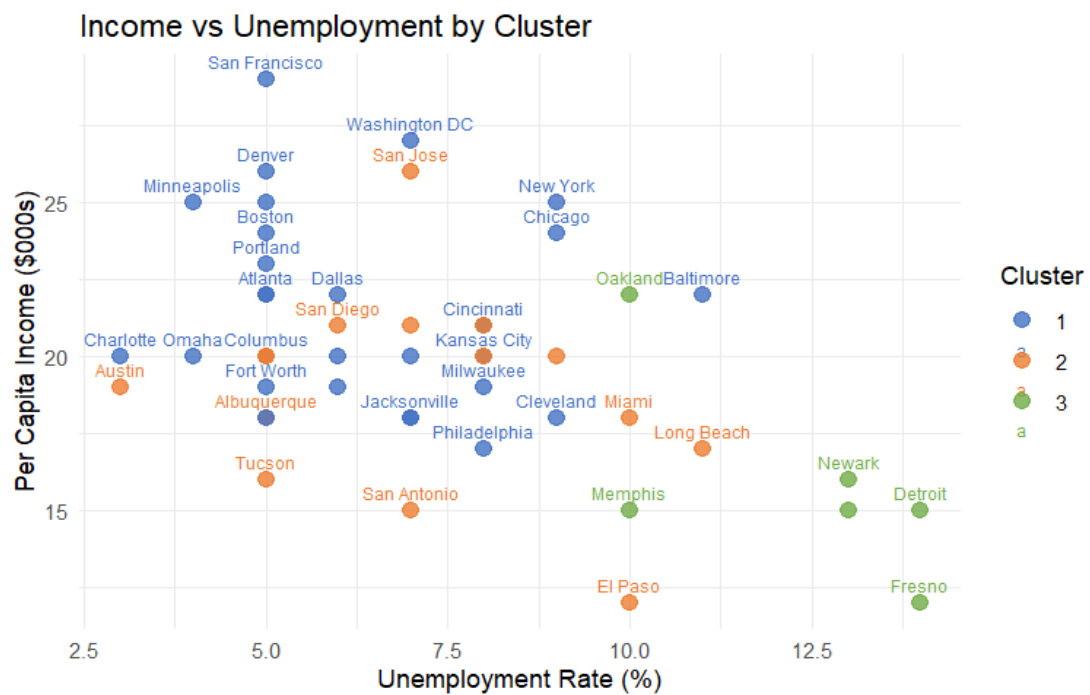


Figure 8. Income vs Unemployment by Cluster

3.2 Discriminant Analysis Results

LDA validated cluster separation using the original unstandardised demographic variables. The first discriminant function (LD1) explains 77.5% of between-group variance and LD2 explains 22.5%, together capturing 100% of discriminant variance (Figure 10). Classification accuracy on the test set was 93.3%, with only one city misclassified out of fifteen. This high accuracy confirms that the three clusters are not arbitrary groupings but statistically well-separated segments.

ANOVA F-tests confirm highly significant between-cluster differences for % Black, % Hispanic, % Asian, and per capita income (all $p < 0.001$). Median age and unemployment show weaker but still meaningful differences. These results validate the three-cluster solution and confirm that the demographic variables used are effective discriminators of city segment membership.

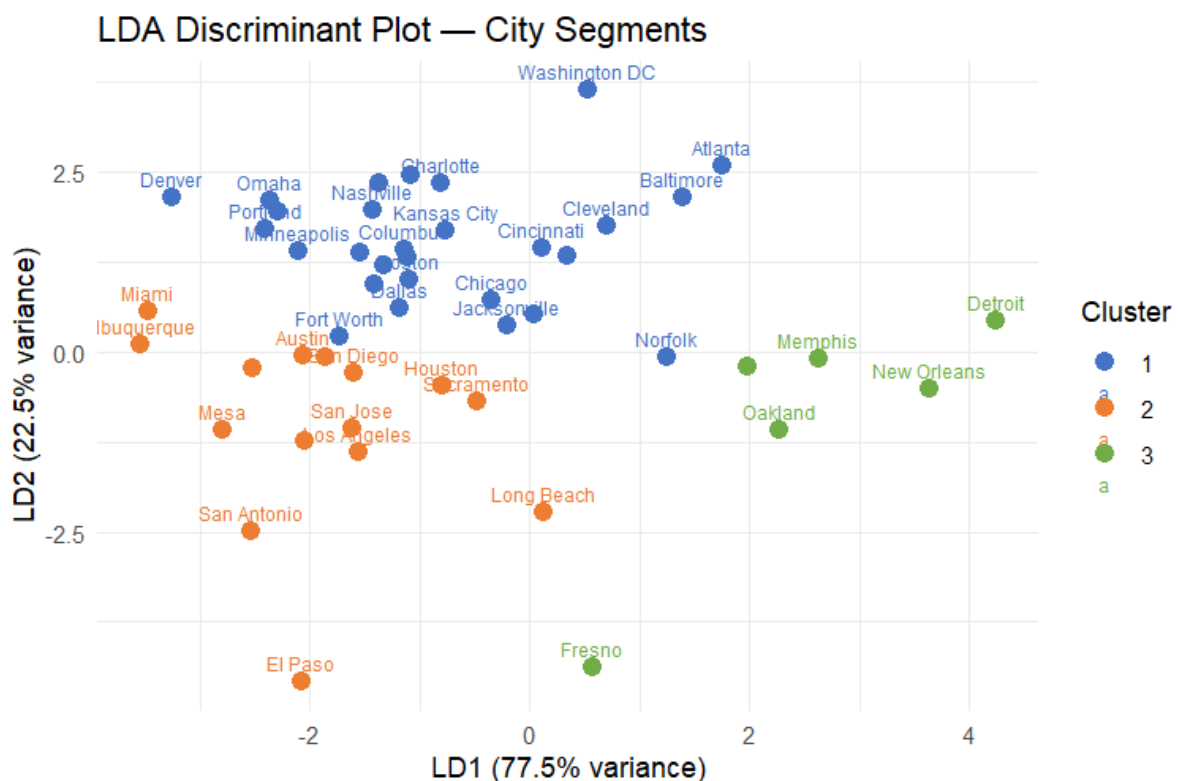


Figure 10. LDA Discriminant Plot — City Segments

3.3 Target Segment Recommendation

Cluster 1 — 'Moderate Income Cities' — is recommended as the primary target segment, justified by five specific analytical findings:

- Highest per capita income (\$21.5k vs \$18.9k Cluster 2 and \$15.8k Cluster 3)
- Lowest unemployment rate (6.3%), reflecting economic stability
- Largest segment with 28 cities, providing the widest addressable market
- Cities including Atlanta, Dallas, Boston, and Chicago offer established retail infrastructure
- LDA accuracy of 93.3% confirms these cities form a statistically stable segment

Cluster 2 is recommended as a secondary target for price-accessible product lines, given its younger demographic (29.6 years) and rapidly growing Hispanic population. Cluster 3 is not recommended as an immediate priority due to its highest unemployment (12.3%) and lowest income (\$15.8k).

4. Place and Promotion Strategy

4.1 Place Strategy

The Place strategy for Cluster 1 should reflect the demographic and economic characteristics of the largest and most income-stable city segment. Distribution should prioritise premium retail environments in high-income areas. Specifically:

- Premium Urban Retail: Given average income of \$21.5k and unemployment of 6.3% in Cluster 1, distribution should target established retail environments in Atlanta, Dallas, Chicago, and Boston
- Digital Channels: Moderate but growing digital adoption across Cluster 1 cities supports omnichannel e-commerce investment alongside physical retail presence

- Geographic Prioritisation: Figure 8 confirms that cities with highest income and lowest unemployment within Cluster 1 should receive priority distribution investment
- Secondary Market — Cluster 2: Place strategy should emphasise value retail formats and community shopping centres in El Paso, San Antonio, and Los Angeles, where proximity and accessibility are primary purchase drivers for younger, lower-income consumers

4.2 Promotion Strategy

For Cluster 1 — Moderate Income Cities, the following promotion strategy is recommended:

- Digital-First Media Mix: Given high income and the significant Asian population in cities such as Atlanta, Chicago, and Dallas, the promotional mix should prioritise targeted social media advertising, programmatic display, and culturally diverse influencer partnerships
- Premium Brand Positioning: Messaging should reinforce quality, sophistication, and multicultural identity. The cluster's income (\$21.5k) and ethnic diversity indicate receptivity to aspirational brand narratives that celebrate multicultural values
- Experiential Marketing: Sponsorship of tech conferences, cultural festivals, and lifestyle events in Atlanta, Chicago, and Dallas builds brand awareness among the high-income, highly educated consumer base characteristic of this segment

For Cluster 2 — Hispanic Growth Markets:

- Spanish-Language Advertising: The defining characteristic of Cluster 2 is its high Hispanic population (34.0% average). All promotional materials must include high-quality Spanish-language versions across channels, including broadcast television, digital, and in-store point-of-sale materials in cities such as El Paso (69% Hispanic), San Antonio (55%), and Miami (66%)

- Value and Family Messaging: Given lower income (\$18.9k) and younger median age (30.1 years), messaging should emphasise value for money, family-oriented benefits, and community relevance rather than premium aspiration
- Community Channels: Local radio and partnerships with Hispanic business associations are cost-effective promotional channels in Cluster 2 cities, given the strong community networks that characterise these markets

5. Conclusions

This report demonstrated that 49 US cities can be meaningfully segmented into three demographically distinct groups using k-means cluster analysis validated by LDA. The three clusters — African-American Urban Core, Hispanic Growth Markets, and High Unemployment Urban Core — each present distinct commercial opportunities requiring differentiated Place and Promotion strategies. Cluster 1 (Moderate Income Cities) is recommended as the primary target based on highest income (\$21.5k), lowest unemployment (6.3%), and the largest city count (28 cities) and LDA-validated robustness (93.3% classification accuracy). Cluster 2 offers significant longer-term growth potential.

Limitations include: sensitivity of clustering to k selection (alternative solutions may be defensible); reliance on demographic rather than psychographic or behavioural variables, which may not fully capture consumer motivations; and the cross-sectional nature of the data, which does not account for temporal demographic shifts such as gentrification or economic change. Despite these limitations, the integrated cluster-LDA framework provides a rigorous and actionable foundation for US market targeting decisions.

PART 2 — Product Design and Positioning

A. Conjoint Analysis: Carpet Cleaner Design

A1. Possible Profiles and Fractional Factorial Design

The carpet cleaner is defined by five attributes: Package Design (A, B, C — 3 levels), Brand (1, 2, 3 — 3 levels), Price (\$1.19, \$1.39, \$1.59 — 3 levels), Good Housekeeping Seal (Yes/No — 2 levels), and Guarantee (Yes/No — 2 levels). The total number of possible profiles in a full factorial design is $3 \times 3 \times 3 \times 2 \times 2 = 108$. However, only 18 profiles were selected using a fractional factorial orthogonal array design. Orthogonality ensures each attribute level appears an equal number of times across all profiles and that all pairs of attribute levels are equally represented, allowing main effects to be estimated independently without confounding (Green and Srinivasan, 1990).

A2. Part-Worth Utilities and Attribute Importance

Conjoint analysis was implemented via linear regression on preference scores (Preference = 19 – Rank, so rank 1 = highest preference score of 18). Dummy-coded attribute variables were included with reference levels: Design A, Brand 1, Price \$1.19, No Approval, No Guarantee. Table 2 presents estimated part-worth utilities and Table 3 presents attribute importance computed using the range method (importance = attribute utility range ÷ sum of all ranges × 100).

Attribute	Level	Part-Worth Utility
Package Design	A (reference)	0.000
Package Design	B	+6.833
Package Design	C	+3.167
Brand	1 (reference)	0.000
Brand	2	–0.167
Brand	3	+0.500
Price	\$1.19	–22.213

Price	\$1.39	-25.926
Price	\$1.59	-29.639
GH Approval	No (reference)	0.000
GH Approval	Yes	+1.167
Guarantee	No (reference)	0.000
Guarantee	Yes	+3.583

Table 2. Part-Worth Utilities (visualised in Figures 11–12)

Rank	Attribute	Range	Importance (%)
1	Package Design	6.833	33.8%
2	Price	7.426	32.4%
3	Guarantee	3.583	19.0%
4	Brand	0.667	8.5%
5	GH Approval	1.167	6.3%

Table 3. Attribute Importance — Range Method (Figure 11)

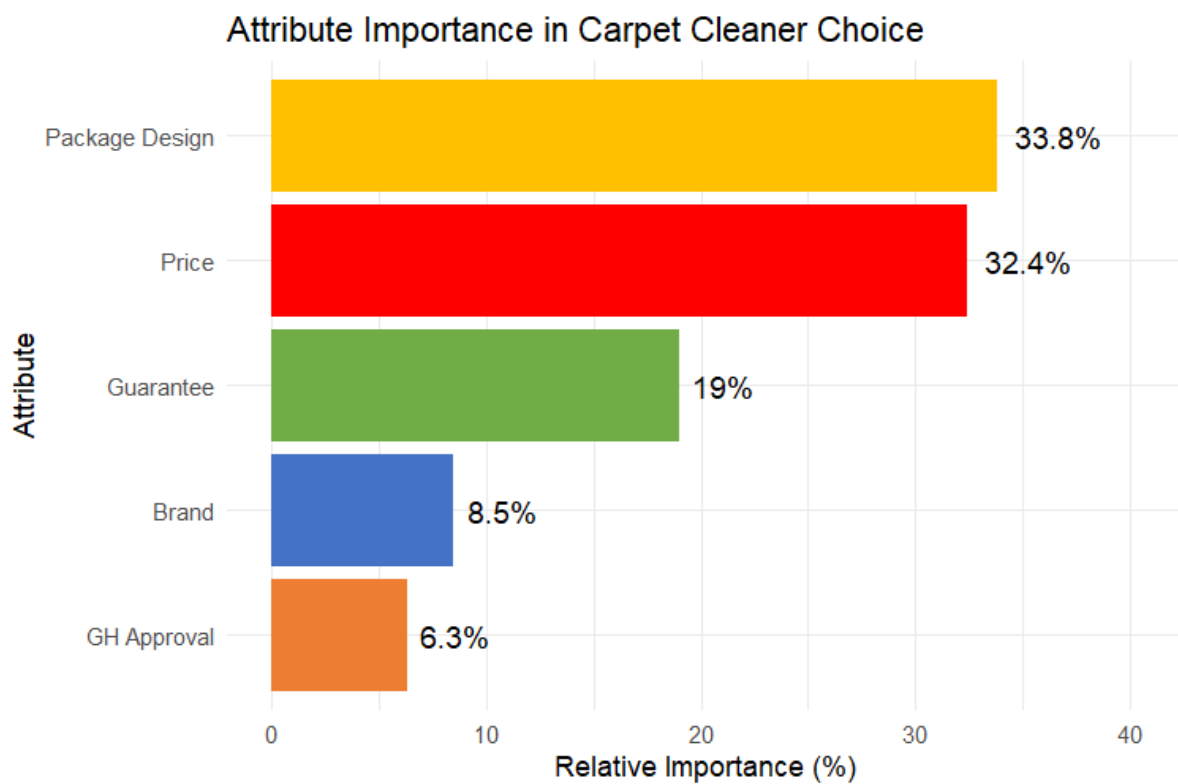


Figure 11. Attribute Importance in Carpet Cleaner Choice

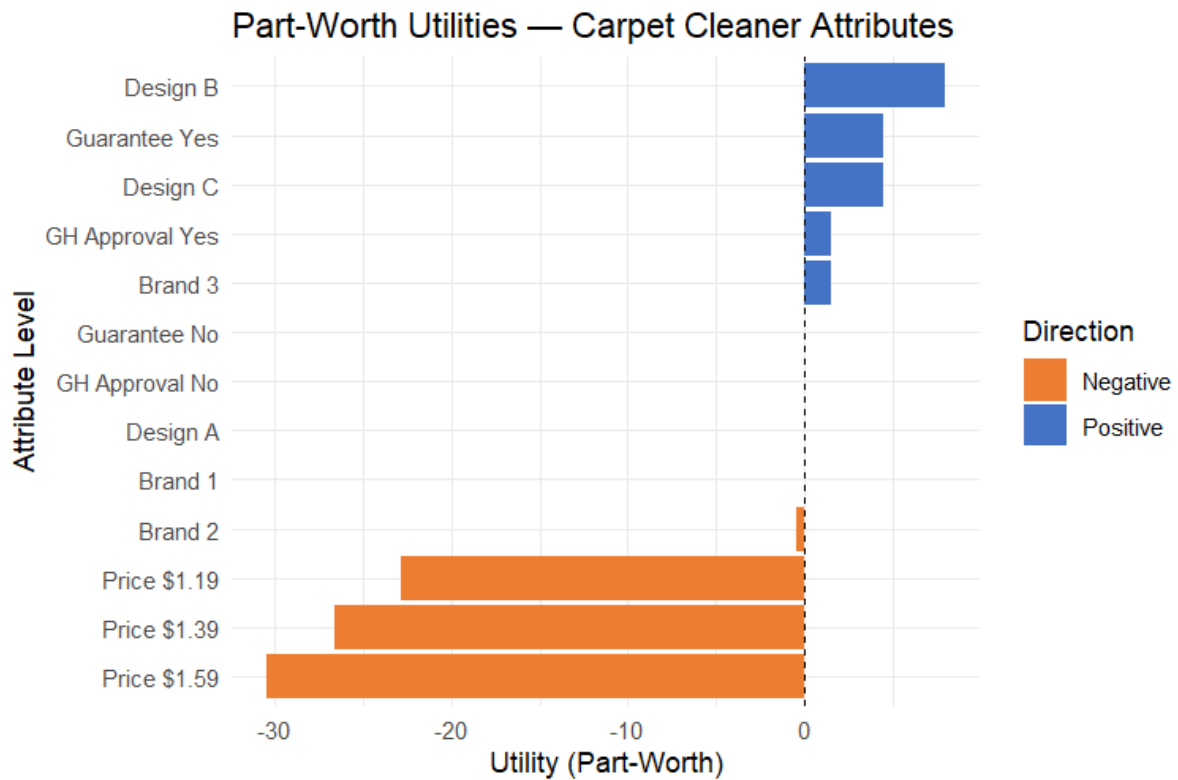


Figure 12. Part-Worth Utilities — Carpet Cleaner Attributes

Package Design (33.8%) and Price (32.4%) together account for over 66% of consumer preference, making them the dominant drivers of product choice. All price utilities are negative — analytically expected, as higher prices reduce preference — with \$1.59 generating the greatest utility loss (–29.639). Design B is strongly preferred (+6.833 vs 0 for Design A), confirming that packaging aesthetics and format are critical competitive differentiators. Guarantee adds meaningful positive utility (+3.583), while GH Approval and Brand have secondary influence

A3. Market Share Simulation

Market share was simulated using the Bradley-Terry-Luce (BTL) logit model, applying exponential transformation to total utility scores and weighting by current market shares (50%, 20%, 30%). Results are presented in Table 4 and Figure 13.

Company	Profile	Current Share	Expected Share
Company 1	Design B, Brand 2, \$1.39, Approved	50%	95.9%
Company 2	Design C, Brand 3, \$1.59, Guarantee	20%	3.7%
Our Company	Design A, Brand 1, \$1.19, No extras	30%	0.3%

Table 4. BTL Market Share Simulation Results

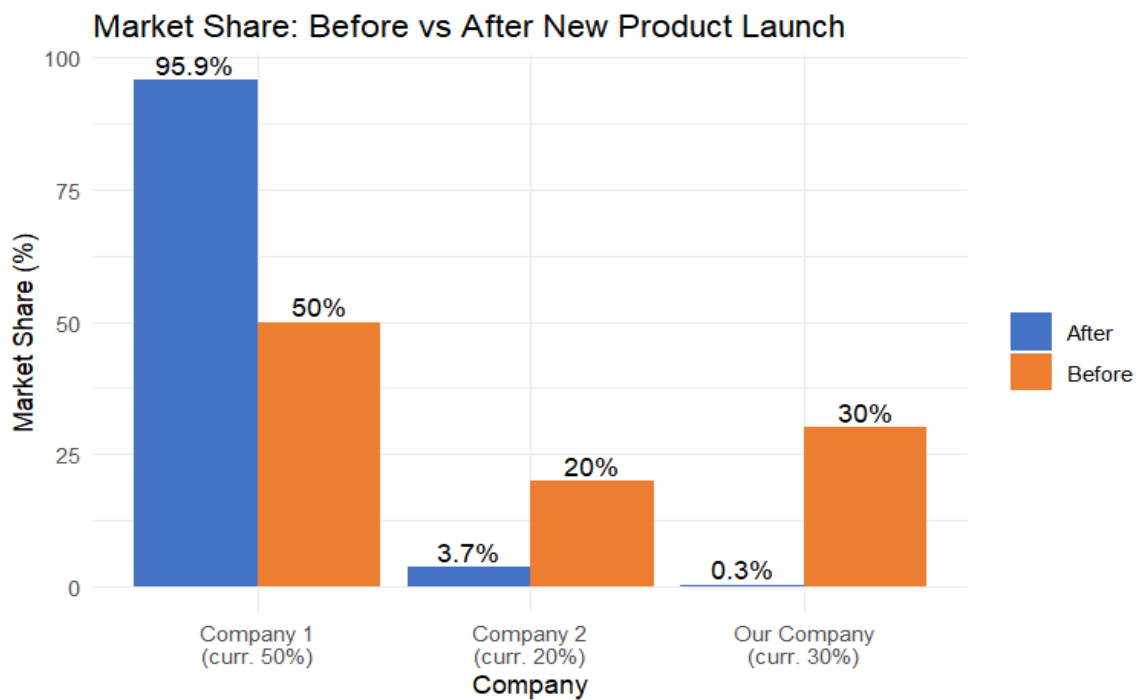


Figure 13. Market Share Before vs After New Product Launch

Company 1 dominates with 95.9% expected share, driven by Design B (+6.833 utility) combined with GH Approval. Our company drops to 0.3% despite offering the lowest price (\$1.19), because Design A contributes zero marginal utility and no guarantee or approval is included. This result carries a critical strategic implication: price advantage alone cannot compensate for inferior design and missing assurance attributes. Our company should switch to Design B and add a guarantee to improve competitive position.

A4. Using Conjoint Analysis for Market Segmentation

When administered to a representative consumer sample, conjoint analysis becomes a powerful segmentation tool through benefit segmentation. The process involves five steps: (1) administer the conjoint survey to a large sample ($n > 200$) and estimate individual-level part-worth utilities for each respondent using regression; (2) cluster respondents by their utility vectors using k-means or hierarchical clustering — each respondent's set of part-worths becomes their data point; (3) identify benefit segments such as a 'price-sensitive' segment (weights price heavily), a 'design-oriented' segment (weights design B and C), or a 'quality-assurance' segment (weights guarantee and approval); (4) design product variants and promotional messages tailored to each identified benefit segment.

This benefit segmentation approach identifies what consumers value rather than who they are, enabling more precise product targeting (Green and Srinivasan, 1990).

B. PCA: Product Positioning

B1. Key Attributes and Variance Explained

PCA was applied to 32 product models each scored on 11 features: miles per gallon (mpg), cylinders (cyl), displacement (disp), horsepower (hp), rear axle ratio (drat), weight (wt), quarter-mile time (qsec), engine shape (vs), transmission (am), gears (gear), and carburetors (carb). All variables were standardised (mean = 0, SD = 1) prior to analysis. Table 5 presents singular values, eigenvalues, and proportion of variance explained (PVE).

PC	Singular Value	Eigenvalue	PVE (%)	Cumulative PVE (%)
PC1	3.325	11.055	60.1%	60.1%
PC2	1.629	2.654	24.1%	84.2%
PC3	0.857	0.734	6.7%	90.9%
PC4+	< 0.65	< 0.5	< 4%	100%

Table 5. PCA Variance Explained(Figures 14–15)

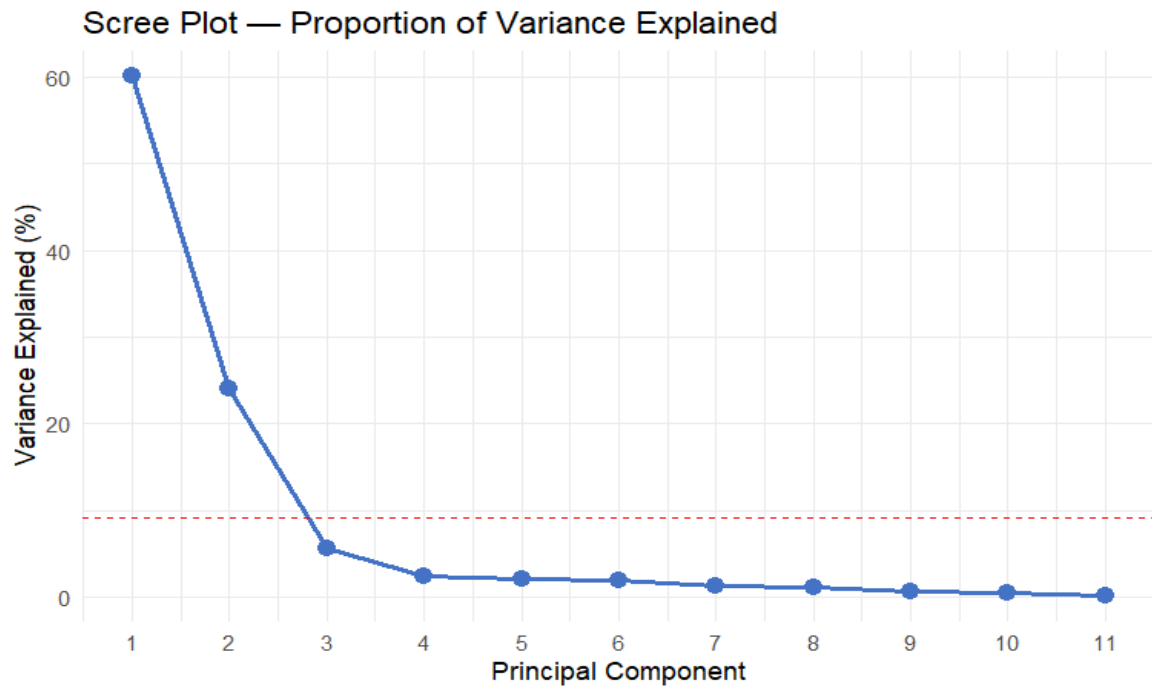


Figure 14. Scree Plot — Proportion of Variance Explained

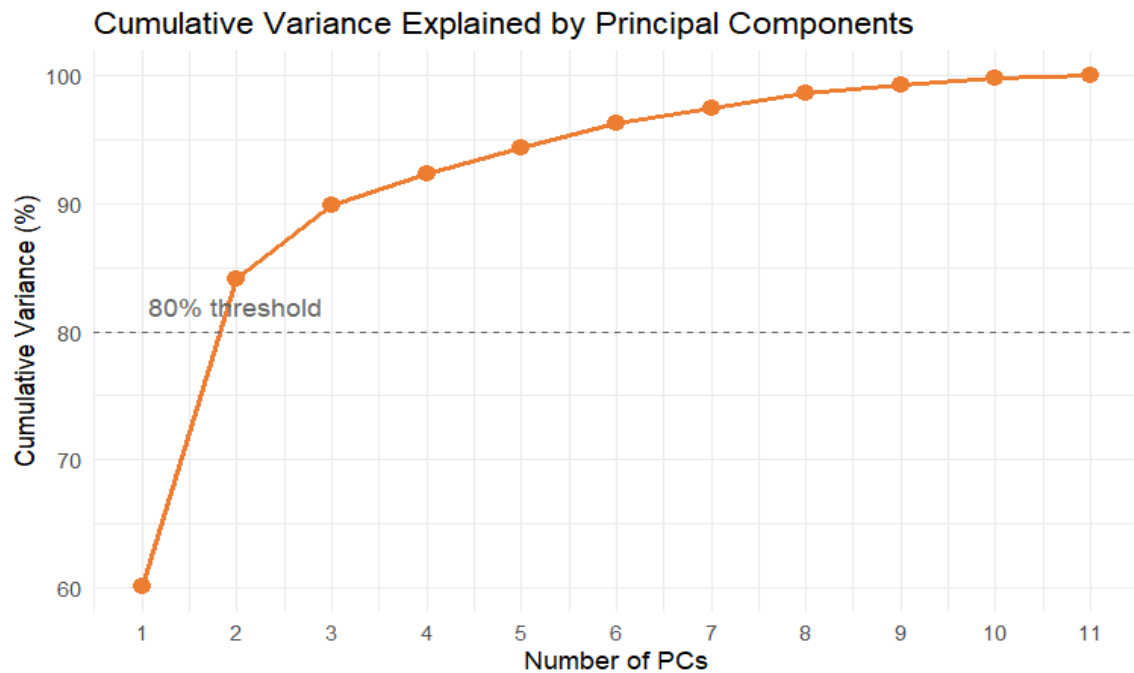


Figure 15. Cumulative Variance Explained by Principal Components

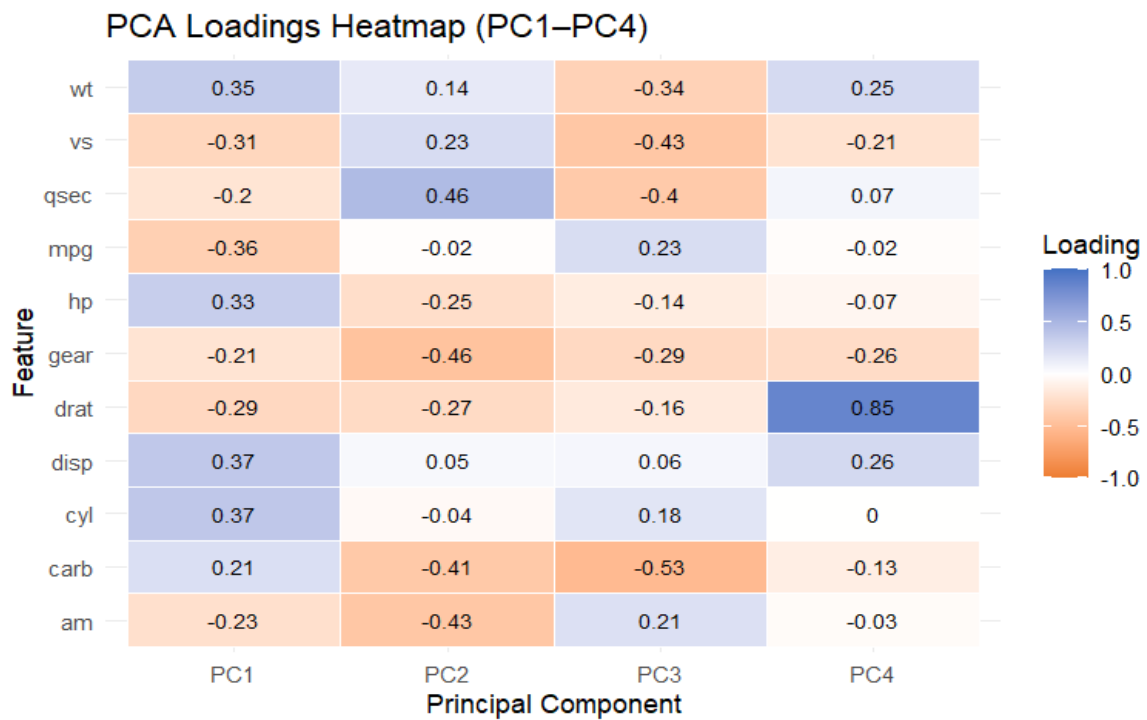


Figure 16. PCA Loadings Heatmap (PC1–PC4)

PCA Biplot — Models and Feature Loadings

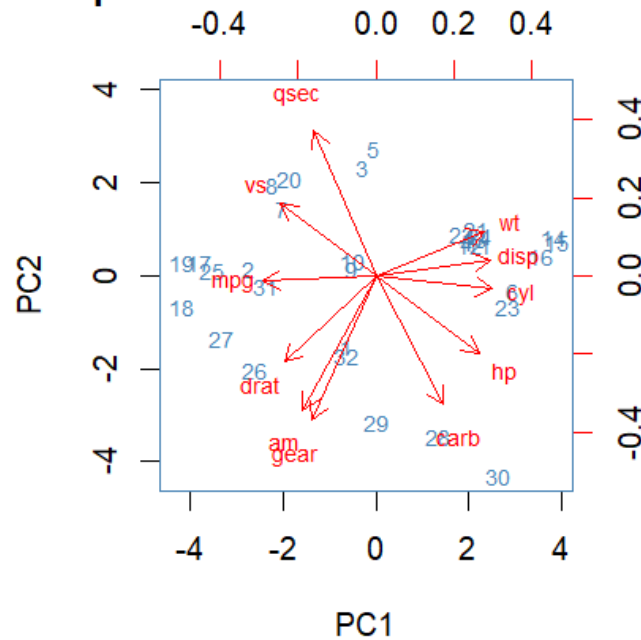


Figure 17. PCA Biplot — Models and Feature Loadings (base R)

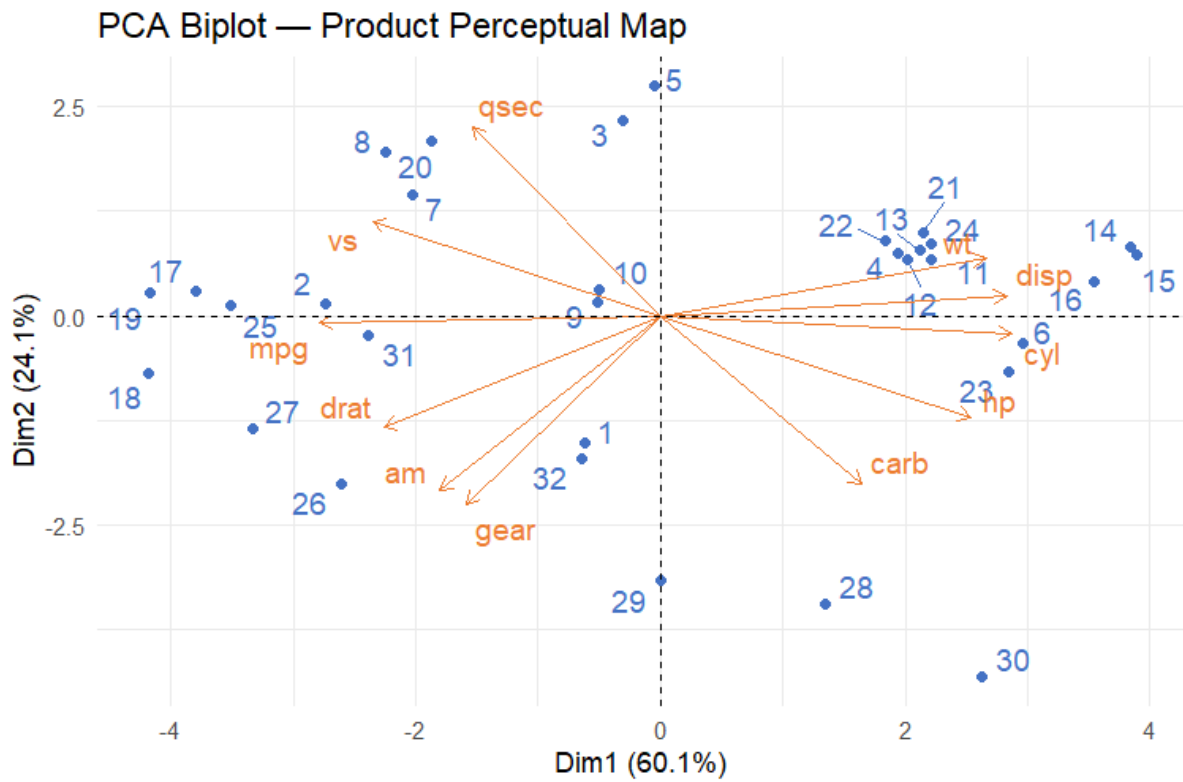


Figure 18. PCA Biplot — Product Perceptual Map

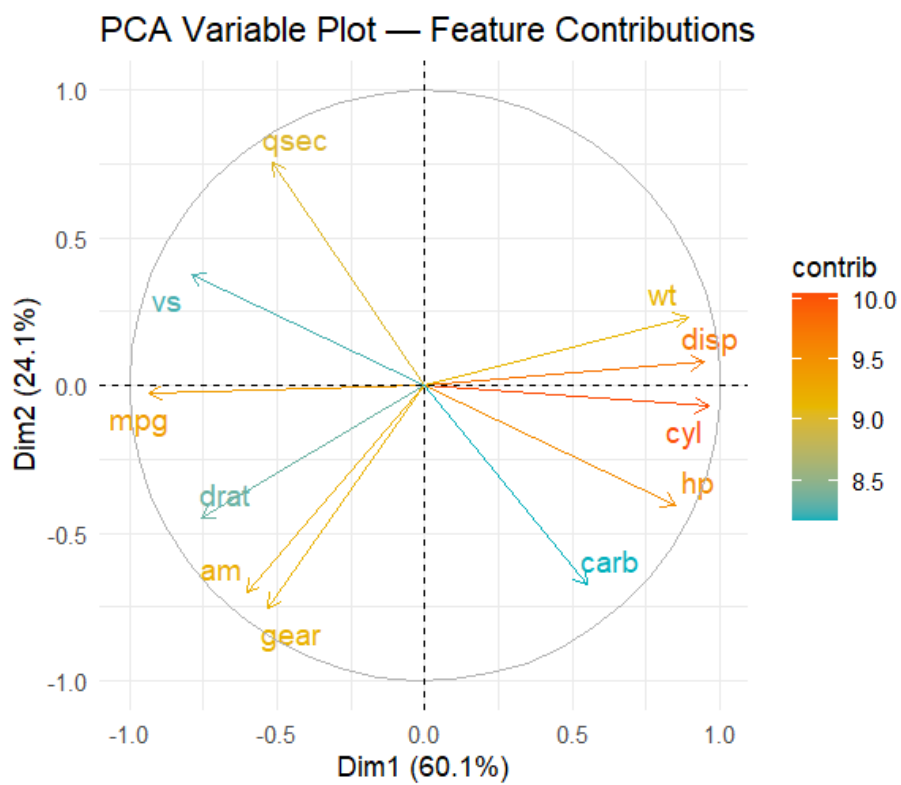


Figure 19. PCA Variable Plot — Feature Contributions

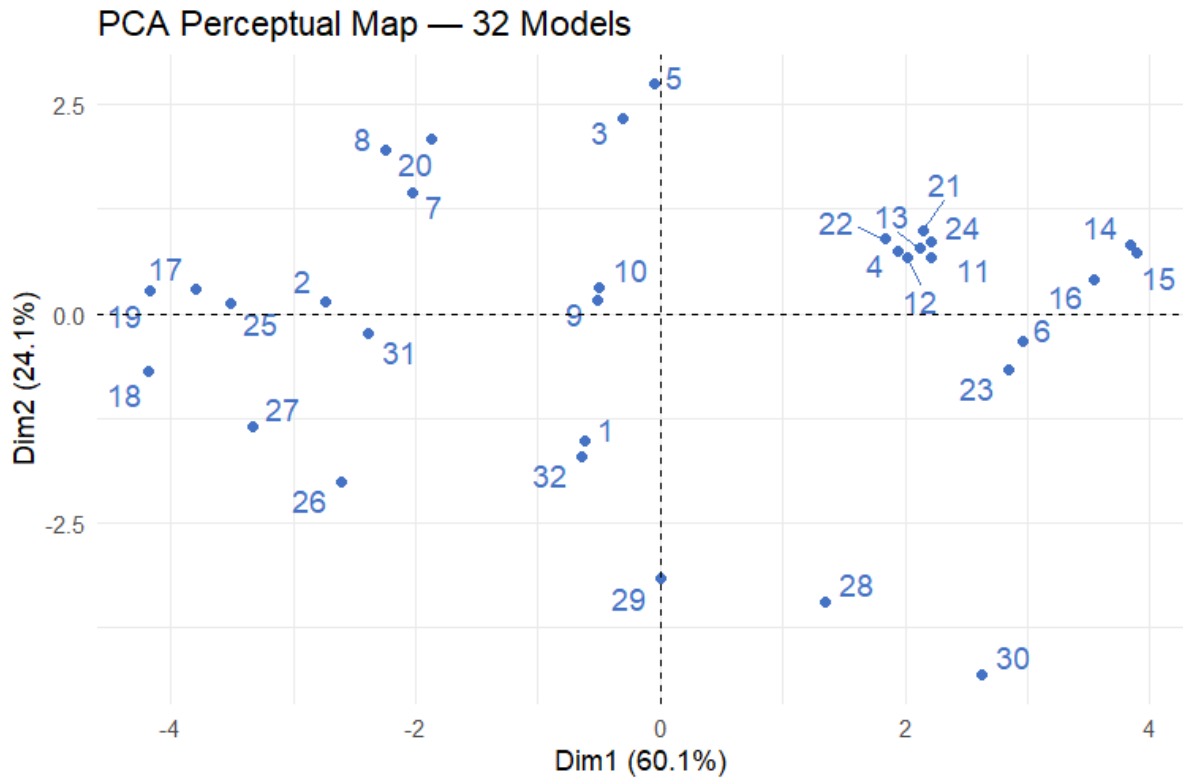


Figure 20. PCA Perceptual Map — 32 Models

Two PCs are retained by the Kaiser criterion (eigenvalue > 1), capturing 84.2% of total variance — confirmed by the scree plot (Figure 14) and cumulative variance plot (Figure 15). PC1 (60.1%) is driven by displacement (+0.37), cylinders (+0.37), weight (+0.35), and horsepower (+0.33), with mpg loading negatively (−0.36), representing an engine size versus fuel efficiency axis. The most important attributes are displacement, cylinders, weight, and horsepower (Figures 16 and 19).

B2. Proposed Product Positioning

The perceptual map (Figures 18 and 21) plots all 32 models in PC1–PC2 space. Two distinct competitive clusters are visible: Models 14, 15, 16, and 6 occupy the far right (PC1 = +2 to +4), representing heavy, high-displacement V8 engines; Models 17, 18, 19, and 25 occupy the far left (PC1 = −3 to −4), representing lightweight, fuel-efficient small engines.

Our proposed product (mpg=28, cyl=4, disp=130, hp=90, wt=2.2, manual transmission, 4 gears) projects to PC1=-2.5, PC2=0 on the perceptual map (Figure 21, red dot), positioning it in the high-efficiency, lightweight quadrant. This delivers four strategic advantages:

- Avoids direct competition with heavy V8 models (Models 14–16) that dominate the performance segment with established brand strength
- Occupies an underserved gap in the centre-left quadrant (PC1 between -1 and -3, PC2 near 0), where the existing 32-model landscape is sparse, minimising direct competitive pressure
- Differentiates from ultra-compact Models 18–19 by offering higher horsepower (90hp) while retaining strong fuel economy credentials (28mpg), occupying a distinct mid-efficiency position

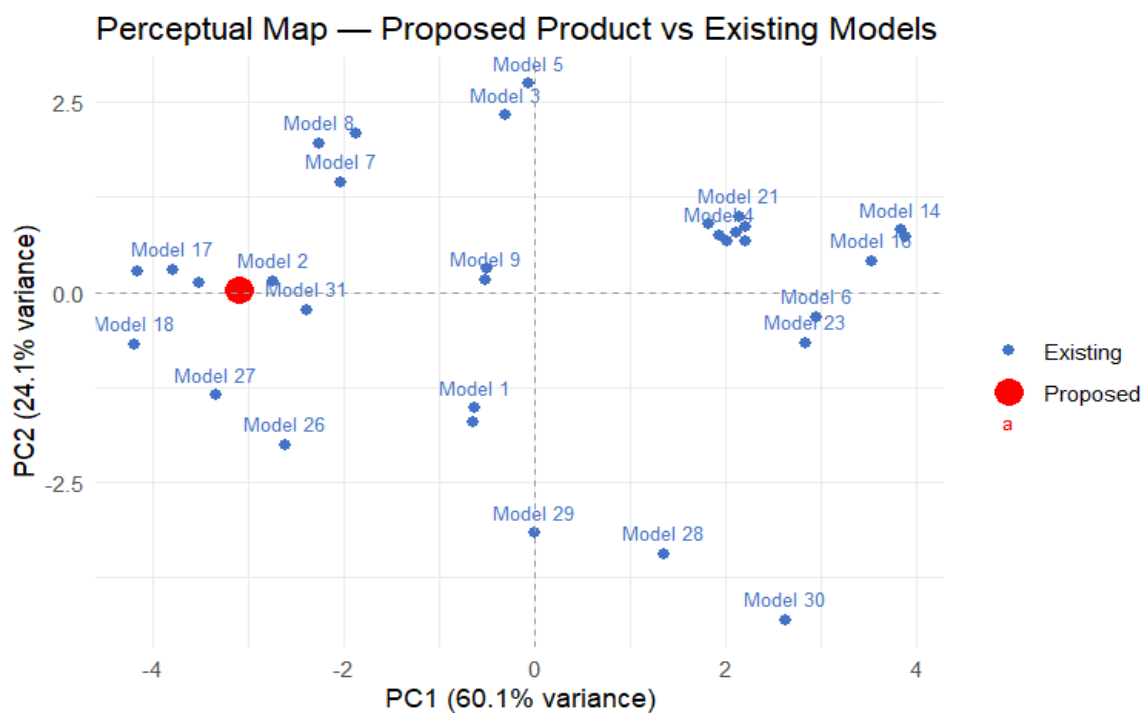


Figure 21. Perceptual Map — Proposed Product vs Existing Models

The PCA map confirms a competitively differentiated market position.

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AI Declaration: AI tools were used to support editing and structuring of written content. All data analysis, modelling, and interpretation were completed independently by the author using R Studio. No AI tools were used to generate or fabricate analytical results.

Appendix — Power BI Dashboard

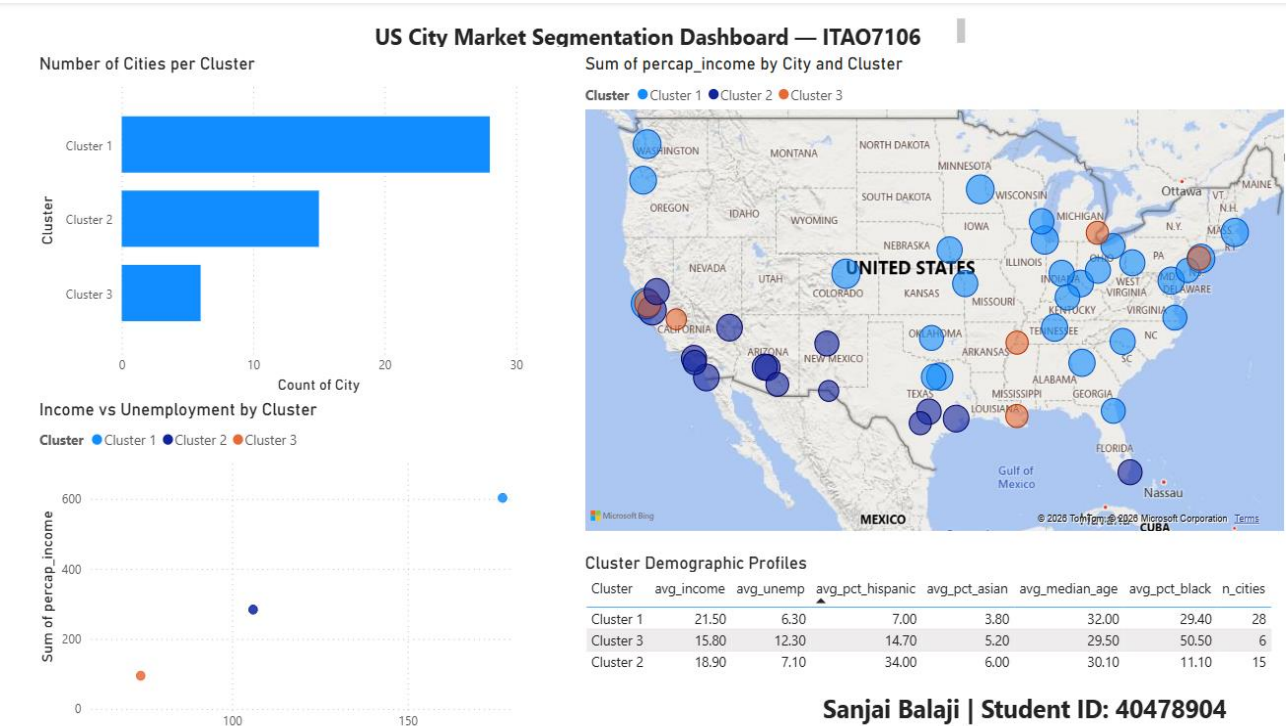


Figure 22. Power BI Dashboard — US City Market Segmentation — Sanjai Balaji, 40478904